

Homework 3 — Solution Notes

Coffee Demand, Conduct, and Counterfactuals

Empirical Industrial Organization, Spring 2026

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This note documents the reference solution and, in particular, the one modification relative to the assignment sheet: in Counterfactual 1 the Peru–Honduras–Mexico bloc coordinates through *partial internalization* $\kappa = 0.5$ rather than a full merger. All objects are produced by `coffee_solution.py`; run `python coffee_solution.py --data coffee.csv --out out/`. Tables and figures are written to `out/`.

1 Setup and notation

Throughout the supply side we use **spec 2** of the linear inverse demand (temperature-only instruments, eq. 1):

$$P_t = \alpha_0 + \alpha_Q Q_t + \alpha_X X_t + \alpha_Z Z_t + u_t \iff P(Q_t; X_t, Z_t) = a_t - b Q_t, \quad a_t \equiv \alpha_0 + \alpha_X X_t + \alpha_Z Z_t, \quad b \equiv -\alpha_Q > 0.$$

$L = \{\text{Brazil, Vietnam, Colombia, Indonesia}\}$ are leaders, the remaining active exporters are followers F . $P_Q = -b$. Operating profit of country i is $\pi_i = (P - c_i)q_i$. Because there is no outside good, inside shares q_i/Q sum to one and consumer surplus is the area of the demand triangle,

$$CS(Q) = \int_0^Q (a_t - bq) dq - P Q = \frac{1}{2} b Q^2, \quad TS = CS + \sum_i \pi_i.$$

The modification to 4(a): partial internalization $\kappa = 0.5$

The bloc members are *followers*, and coffee is a *homogeneous* good. Both facts matter. The follower first-order condition (Appendix eq. 7) with a follower ownership matrix Ω^F is

$$a_t - bQ - c_f - b \sum_{k \in F} \Omega_{fk}^F q_k = 0. \quad (1)$$

The assignment's "merger" sets $\Omega_{ij}^F = 1$ for the three bloc members. In a homogeneous good this is degenerate: every cross inverse-demand derivative equals $P_Q = -b$, so internalizing a bloc partner is *identical* to internalizing oneself. Differencing (1) for two bloc members i, j gives

$$b[(1 + \Omega_{ii}^F)q_i - \Omega_{ij}^F q_j] - b[(1 + \Omega_{jj}^F)q_j - \Omega_{ji}^F q_i] = c_j - c_i \implies b(1 - \kappa)(q_i - q_j) \text{ (within-bloc)} = -(c_i - c_j).$$

At $\kappa = 1$ the left-hand side collapses to $0 = -(c_i - c_j)$: with heterogeneous recovered costs there is *no interior split*—the bloc wants to run only its cheapest member—and the matrix $\mathbf{1}\mathbf{1}' + \Omega^F$ is singular, so the equilibrium is not defined (numerically the solver only survives via a pseudo-inverse, which returns an economically meaningless minimum-norm allocation). With $\kappa = 0.5$ the within-bloc output split is finite and determinate,

$$q_i - q_j = -\frac{c_i - c_j}{b(1 - \kappa)},$$

so all three members stay interior and the object $\Delta\Pi_t^{PHM}$ is the value of *partial* coordination (the maximum the bloc would jointly pay to reach 50% internalization), not the value of fusing into a single firm. This is both what was requested and the better-posed object. Concretely, Ω^F is the identity except for the 3×3 bloc, whose off-diagonal entries are set to $\kappa = 0.5$.

Caveat used in grading. The joint gain $\Delta\Pi_t^{PHM} > 0$ need not make cooperation individually rational for every member: the highest-cost bloc member is asked to cut output the most and can see its *own* profit fall, so the coordination is sustained only with internal transfers. Report each member's individual $\Delta\Pi$ and flag this.

2 Demand (Section 2)

Signs. Expect $\alpha_Q < 0$ (inverse demand slopes down), $\alpha_X > 0$ (higher importer GDP shifts demand out, raising price at given Q), and $\alpha_Z > 0$ (tea is a substitute: a higher tea price pushes buyers toward coffee). **OLS vs. IV.** Q_t is simultaneously determined, so OLS is biased: supply shocks trace out the demand curve and attenuate $|\hat{\alpha}_Q|$ (often pushing it toward zero). The cost-shifter instruments (fertilizer, temperature) move quantity through supply only, restoring a steeper, more negative $\hat{\alpha}_Q$. Spec 1 (fert + temp) is more efficient if fertilizer is valid; spec 2 (temp only) is the more defensible exclusion restriction and is used downstream.

Elasticities. For the linear form $\varepsilon_t = \frac{\partial Q}{\partial P} \frac{P_t}{Q_t} = \frac{1}{\alpha_Q} \frac{P_t}{Q_t}$ (time-varying); for the log form $\varepsilon = 1/\beta_Q$ (constant). Coffee demand is typically *inelastic* ($|\varepsilon| < 1$) over most of the sample, which is what makes the market-power story economically interesting. Report `demand_estimates.csv` and `elasticities.png`.

3 Marginal cost and markups (Section 3)

Given b and a_t , evaluate the conduct FOCs at observed (P_t, q_{it}) .

(a) Cournot. Every country's FOC is $P + P_Q q_i = c_i$, hence

$$c_{it}^{\text{Cournot}} = P_t - b q_{it}, \quad \text{markup} = P_t - c_{it} = b q_{it}. \quad (2)$$

(b) Stackelberg. Followers face Cournot competition among themselves given the leaders' output, so their FOC—and therefore their recovered cost—is *identical* to Cournot: $c_f = P_t - b q_{ft}$. Leaders anticipate follower accommodation. With N_F symmetric-slope followers and $\Omega^F = I$, the total pass-through of leader output into Q is

$$\tau = 1 - \mathbf{1}'(\mathbf{1}\mathbf{1}' + \Omega^F)^{-1}\mathbf{1} = \frac{1}{1 + N_F},$$

and the leader FOC (Appendix eq. 8) $P - c_\ell + P_Q q_\ell \tau = 0$ gives

$$c_{\ell t}^{\text{Stackelberg}} = P_t - b q_{\ell t} \tau, \quad \text{markup} = b q_{\ell t} \tau. \quad (3)$$

Interpretation (the required discussion). Only the four leaders differ across the two conduct assumptions; all followers coincide. Because $\tau < 1$, the Stackelberg model attributes a *smaller* markup and hence a *higher* marginal cost to each leader than Cournot does: with a large accommodating fringe ($\tau \approx 1/(1 + N_F)$ is small), a leader's residual demand is nearly flat, its first-mover power is weak, and the same observed output is rationalized only by a cost close to price. The per-leader panels `mc_markup_{\{Brazil,Vietnam,Colombia,Indonesia\}.png` show this gap.

4 Counterfactuals (Section 4)

Equilibrium solver. With linear demand the two stages are linear. Given leaders' aggregate Q_L , followers solve $b(\mathbf{1}\mathbf{1}' + \Omega^F)q_F = (a_t - bQ_L)\mathbf{1} - c_F$, i.e. $q_F(Q_L) = A - BQ_L$ with $A = \frac{1}{b}M^{-1}(a_t\mathbf{1} - c_F)$, $B = M^{-1}\mathbf{1}$, $M = \mathbf{1}\mathbf{1}' + \Omega^F$, and $\tau = 1 - \mathbf{1}'B$. Leaders then solve $b\tau(\mathbf{1}\mathbf{1}' + I)q_L = (a_t - b\mathbf{1}'A)\mathbf{1} - c_L$. Aggregate $Q = Q_L + \mathbf{1}'q_F$, $P = a_t - bQ$, profits, $CS = \frac{1}{2}bQ^2$, and TS follow. Marginal costs are the Stackelberg-recovered $c^{\text{Stackelberg}}$ with the negative-cost floor (replace $c < 0$ by the 25th percentile of that year's non-negative costs).

Non-negativity (shut-down) rule. If the linear solution returns a negative quantity for some country, that country does not produce: set $q_i = 0$, remove it from the game, and re-solve the reduced equilibrium. Iterate until every active country has $q \geq 0$. Since removing a producer raises the price and weakly raises every survivor's output, the iteration drops the most-negative country each round and terminates in at most N steps at the unique cost-consistent active set (every active country has $c_i < P^*$, every shut-down country has $c_i \geq P^*$, so none would re-enter). A shut-down country earns zero operating profit. The count of shut-down countries is reported per counterfactual.

Consequence for comparative statics. Because entry/exit is now a discrete margin, some counterfactual responses can be non-monotone: a shock that raises the price can pull a previously-shut country back in, and its entry partially offsets the price move. Read the signs below as the interior mechanism; where the shut-down margin binds, take the reported table as authoritative.

CF1 — Value of (partial) cooperation, $\kappa = 0.5$. Set the bloc block of Ω^F to 0.5. The bloc restricts output, P rises, and $\Delta\Pi_t^{PHM} = (\pi^{Peru} + \pi^{Hond} + \pi^{Mex})_{\kappa=0.5} - (\cdot)_{\text{base}} > 0$ measures the joint value of coordination. Report each member's $\Delta\Pi$ in absolute and as % of base profit; "who would pay more" is the member with the largest percentage gain. Expect the answer to be redistributive: the member asked to cut most may lose individually (see the caveat above).

CF2 — Bad crop among leaders (+5% leader MC). Leaders contract; through τ this raises P . **Followers benefit:** they face the same higher price with unchanged costs and expand along their reaction functions—the classic leader-follower free-riding result. Leader profits fall; follower profits rise; CS and TS fall. The mechanism is that leaders bear the cost shock while the price increase they induce is a public good the fringe captures.

CF3 — Tea more competitive ($Z \rightarrow 0.95Z$). Only the demand intercept moves: $\Delta a_t = \alpha_Z(0.95 - 1)Z_t < 0$, shifting demand in. Both P and Q fall, so every producer loses; decompose each country's loss into the price and quantity channels. Leaders lose more in absolute terms (they are larger); the percentage ranking is data-dependent—report it from `table_CF3.csv` rather than asserting it.

CF4 — Value of leadership. Move Indonesia from L to F and re-solve. $\Delta\Pi_t^{\text{Indonesia}} = \pi^{\text{Indonesia, follower}} - \pi^{\text{Indonesia, leader}} < 0$ is the strategic value of moving first (the first-mover/Stackelberg rent Indonesia forgoes). Effects on the remaining leaders and the fringe are ambiguous—removing a leader raises the survivors' residual market power but adds Indonesia's output to the accommodating fringe—so report signs from `table_CF4.csv`.

Deliverable tables/figures. `table_CF{1,2,3,4}.csv` (average P , total Q , CS , leader/follower profit, TS , count of shut-down countries) and `profit_changes.png` (country-level % profit change across all four counterfactuals).

A Symbol-to-code map

Object	Code
$a_t = \alpha_0 + \alpha_X X_t + \alpha_Z Z_t$	<code>a_t in recover_costs/run_counterfactuals</code>
$b = -\alpha_Q$ (spec 2)	<code>b; from demand["lin_IV2"]</code>
$\tau = 1/(1 + N_F)$	<code>returned by follower_block(...)</code>
$c^{\text{Cournot}}, c^{\text{Stackelberg}}$	<code>mc_cournot, mc_stack in marginal_costs.csv</code>
Negative-cost floor	<code>floor_costs()</code>
Ω^F with $\text{bloc} = \kappa$	<code>build_OmegaF(followers, BLOC, KAPPA)</code>
Forward equilibrium (shut-down on)	<code>solve_stackelberg(a,b,mc,leaders,followers,bloc,kappa)</code>
Single linear solve (shut-down off)	<code>_solve_linear(...) / nonneg=False</code>
Shut-down set	<code>eq["shutdown"]; count in table_CF*.csv</code>
$CS = \frac{1}{2}bQ^2, TS$	<code>inside solve_stackelberg</code>

Implementation note. The shut-down rule is implemented as the active-set iteration described above (drop the most-negative country, re-solve, repeat). Removal order is by most-negative quantity; the resulting set is the unique one with all active $c_i < P^*$ and all shut-down $c_i \geq P^*$, so the solution does not depend on the order in which ties are broken. The only borderline case is a country whose equilibrium quantity is essentially zero; treating it as in (producing ≈ 0) or out gives the same price and welfare to numerical tolerance.